# Module 3

- The key to analysing and understanding the performance of digital transmission is the realization that
- signals used in communications can be expressed and visualized graphically
- Thus, we need to understand signal space concepts as applied to digital communications

# Traditional Bandpass Signal Representations

- Baseband signals are the message signal generated at the source
- Passband signals (also called bandpass signals) refer to the signals after modulating with a carrier. The bandwidth of these signals are usually small compared to the carrier frequency fc
- Passband signals can be represented in three forms
  - Magnitude and Phase representation
  - Quadrature representation
  - Complex Envelop representation

# Magnitude and Phase Representation

$$s(t) = a(t)\cos\left[2\pi f_c t + \theta(t)\right]$$

where $a(t)$ is the magnitude of $s(t)$ and $\theta(t)$ is the phase of s(t)

# Quadrature or I/Q Representation

$$s(t) = x(t)\cos(2\pi f_c t) - y(t)\sin(2\pi f_c t)$$

where x(t) and y(t) are real-valued baseband signals called the in-phase and quadrature components of s(t).
Signal space is a more convenient way than I/Q representation to study modulation scheme

# Model of Transmitter

- We consider the following model of a generic transmission system (*digital source*):
  - A message source transmits 1 symbol every T sec
  - Symbols belong to an alphabet M ($m_1$, $m_2$, …$m_{M)}$
    - Binary – symbols are 0s and 1s
    - Quaternary PCM – symbols are 00, 01, 10, 11

- Symbol generation (message) is probabilistic, with a priori probabilities $p_1$, $p_2$, .. $p_M$. or
- Symbols are equally likely
- So, probability that symbol $m_i$ will be emitted:

- Transmitter takes the *symbol (data)* $\boldsymbol{m_i}$ (digital message source output) and encodes it into a *distinct signal* $\boldsymbol{s_i(t)}$.
- The *signal* $s_i(t)$ occupies the whole ***slot T*** allotted to *symbol* $m_i$.
- $s_i(t)$ is a real valued energy signal (signal with finite energy)

- Task of transforming an incoming message $m_i$, into i= 1,2 ,3 ,4 ,5 ……M into modulated wave Si(t) may be divided into separate discrete-time and continuous time operations.

# Geometric Representation of Signals

- Objective: To represent any set of ***M*** *energy signals* *$\{s_i(t)\}$* as linear combinations of ***N*** *orthogonal basis functions*, where N ≤ M
- Real value energy signals *$s_1(t)$, $s_2(t)$,..$s_M(t)$,* each of duration T sec

Orthogonal basis function

$$s_i(t) = \sum_{j=1}^{N} s_{ij}\phi_j(t), \quad \begin{Bmatrix} 0 \le t \le T \\ i==1,2,....,M \end{Bmatrix}$$

coefficient

Energy signal

Coefficients of expansion are

$$s_{ij} = \int_0^T s_i(t)\phi_j(t)dt, \qquad \begin{Bmatrix} \text{i=1,2,....,M} \\ \text{j=1,2,....,M} \end{Bmatrix}$$

Real-valued basis functions $\phi_1(t)$, $\phi_2(t)$, … $\phi_N(t)$ are orthonormal …which means

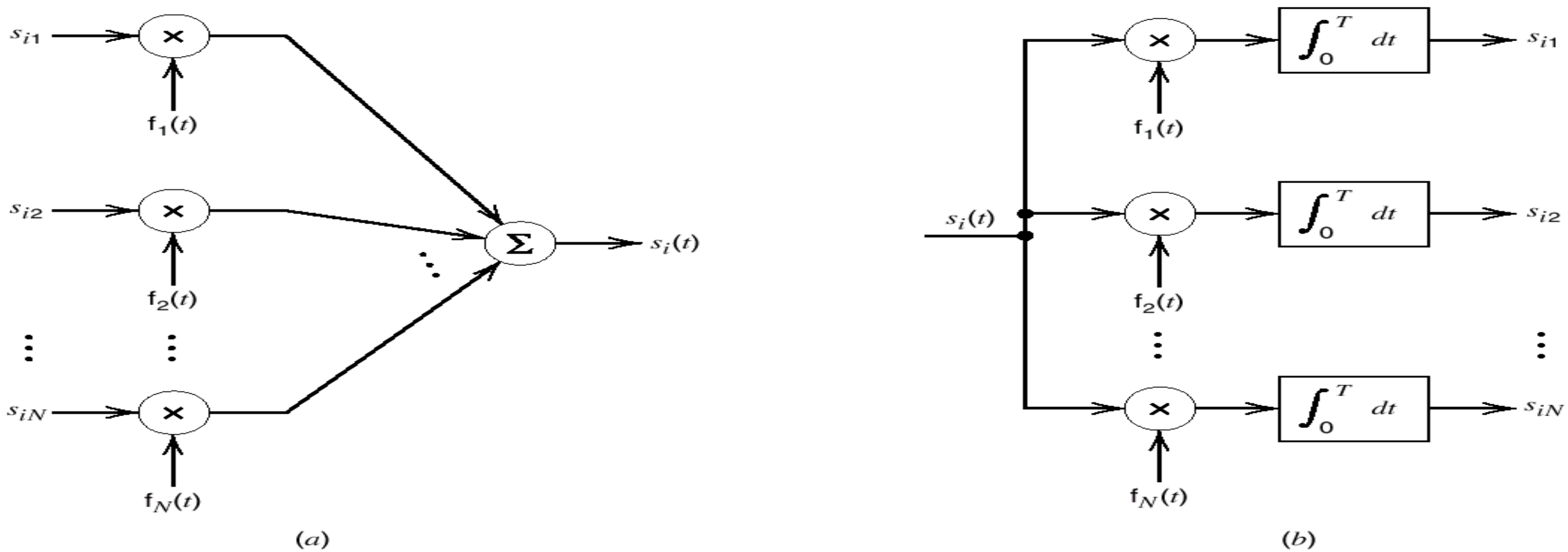
$$\int_0^T \phi_i(t)\phi_j(t)dt = \delta_{ij} = \begin{Bmatrix} 1 \ \text{ if } i = j \\ 0 \ \text{ if } i \neq j \end{Bmatrix}$$

Each basis function is normalised to have unit energy.

basis functions $\phi_1(t)$, $\phi_2(t)$, … $\phi_N(t)$ are orthogonal With respect to each other.

- The set of coefficients can be viewed as a N-dimensional vector, denoted by $s_i$
- Bears a one-to-one relationship with the transmitted signal $s_i(t)$

(a)Scheme for generating the signal $s_i(t)$.
(b)Scheme for generating the set of signal vectors $\{s_i\}$




can be viewed as modulator and detector

# Stage 1

- Establish whether or not given set of signals, $s_1(t)$, $s_2(t),..s_M(t)$ is linearly independent.
- $a_1 s_1(t) + a_2 s_2(t) \ldots\ldots + a_M s_M(t) = 0$

*We may express $s_M(t)$ as*

$$s_M(t) = \frac{a_1}{a_M} s_1(t) + \frac{a_2}{a_M} s_2(t) + \ldots \ldots + \frac{a_{M-1}}{a_M} s_{M-1}(t)$$

*which implies that $s_M(t)$ can be expressed in terms of remaining M-1 signals.*

- Consider set of signals, $s_1(t), s_2(t),..s_{M-1}(t)$.
- *Is this set of signals linearly independent?*
- *If not , there exists a set of numbers $b_1, b_2, ..b_M$-1 such that*
- $b_1 s_1(t) + b_2 s_2(t) \ldots\ldots + b_{M-1} s_{M-1}(t) = 0$

*We may express $s_{M-1}(t)$ as*

$$s_{M-1}(t) = \frac{b_1}{b_{M\_1}} s_1(t) + \frac{b_2}{b_{M\_1}} s_2(t) + \ldots \ldots + \frac{b_{M-2}}{a_{M\_1}} s_{M-2}(t)$$

*Continue…..*

*We will end up with linearly independent subset of original signals.*

*Let $s_1(t), s_2(t),..s_N(t)$ denote this subset.*

# Stage 2

- Show that it is possible to construct a set of N orthonormal basis function $\phi_1(t)$, $\phi_2(t)$, … $\phi_N(t)$

Assume a set of M energy signals denoted by $s_1(t)$, $s_2(t)$, .. , $s_M(t)$.

1. Define the first basis function starting with $s_1$ as: (where E is the energy of the signal)
2. Then express $s_1(t)$ using the basis function and an energy related coefficient $s_{11}$ as:
3. Later using $s_2$ define the coefficient $s_{21}$ as:

$$\phi_1(t) = \frac{s_1(t)}{\sqrt{E_1}}$$

$$s_1(t) = \sqrt{E_1}\phi_1(t) = s_{11}\phi_1(t)$$

$$s_{21} = \int_0^T s_2(t)\phi_1(t)dt$$

4. If we introduce the intermediate function $g_2$ as:

$$g_2(t) = s_2(t) - s_{21}\phi_1(t)$$

Orthogonal to $\varphi_1(t)$

5. We can define the second basis function $\phi_2(t)$ as:

$$\phi_2(t) = \frac{g_2(t)}{\sqrt{\int_0^T g_2^2(t)dt}}$$

6. Which after substitution of $g_2(t)$ using $s_1(t)$ and $s_2(t)$ it becomes:

$$\phi_2(t) = \frac{s_2(t) - s_{21}\phi_1(t)}{\sqrt{E_2 - s_{21}^2}}$$

- Note that $\phi_1(t)$ and $\phi_2(t)$ are orthogonal that means:

$$\int_0^T \phi_2^2(t)dt = 1$$

$$\int_0^T \phi_1(t)\phi_2(t)dt = 0$$

# And so on for N dimensional space…,

- In general a basis function can be defined using the following formula:

$$g_i(t) = s_i(t) - \sum_{j=1}^{i-1} s_{ij} \text{-} \phi_j(t)$$

- where the coefficients can be defined using:

$$s_{ij} = \int_0^T s_i(t)\phi_j(t)dt \quad , \qquad j = 1, 2, \ldots.., i-1$$

- Explain the Gram-Schmidt procedure?
  1. Choose one vector, normalize it → first basis.
  2. Project next vector onto current basis. If projection error is zero, no new basis. Otherwise normalize projection error (orthogonal to current basis) → next basis.
  3. Repeat step 2 for all remaining vectors.

- Consider the four signals s1(t), s2(t), s3(t) and s4(t) as shown in the fig-P1.1 . Use Gram-Schmidt Orthogonalization Procedure to find the orthonormal basis for this set of signals. Also express the signals in terms of the basis functions.

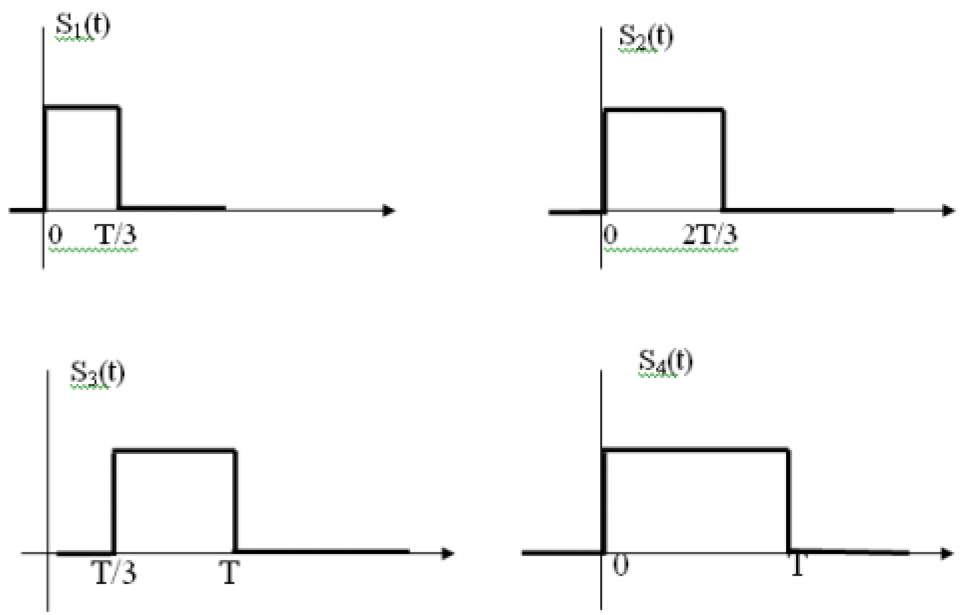



*Fig-P1.1: Signals for the problem -1.*

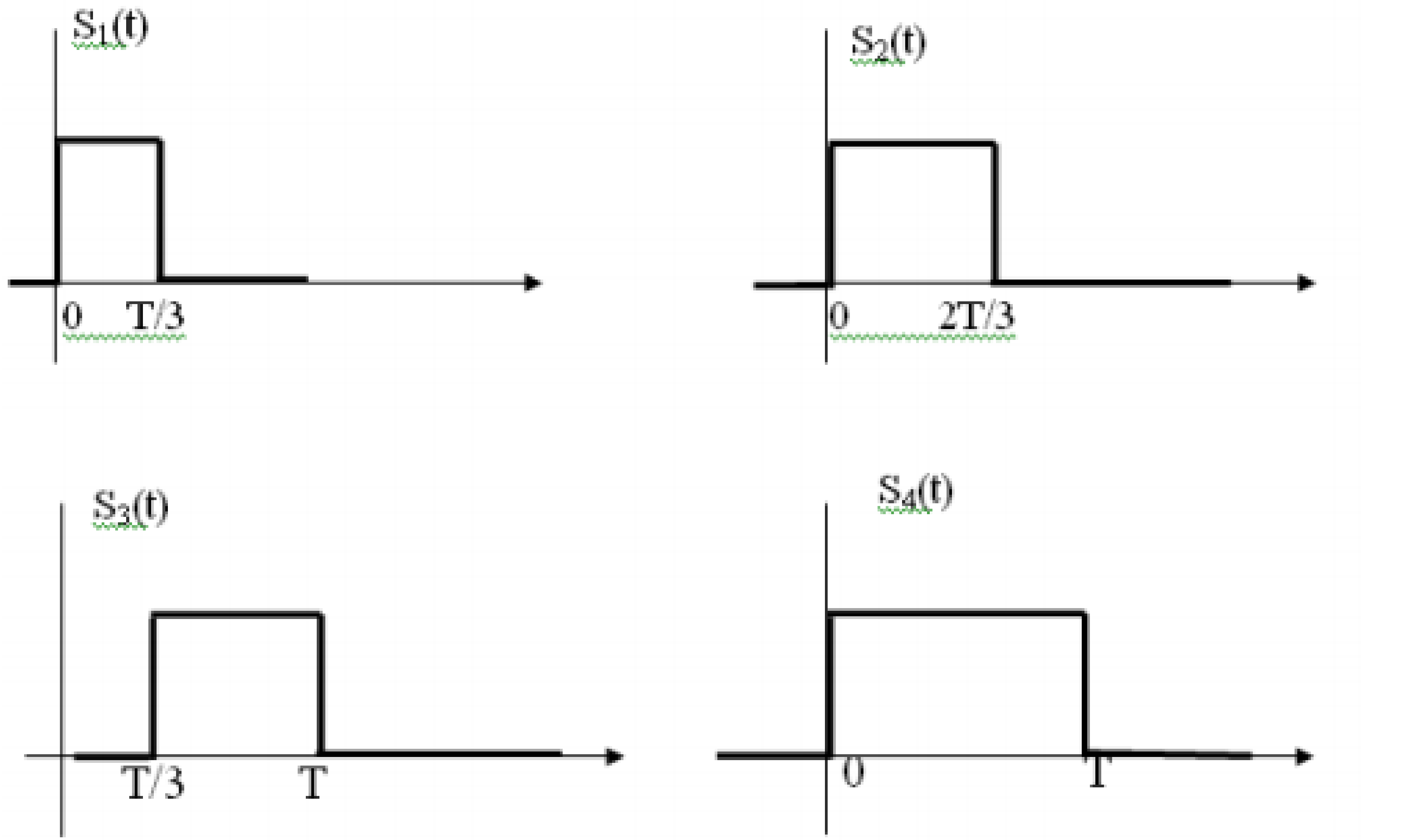



*Fig-P1.1: Signals for the problem -1.*

Given set is not linearly independent because $s_4(t) = s_1(t) + s_3(t)$

Step-1 : Energy of the signal $s_1(t)$

$$E_1 = \int_0^T s_1^2(t)\, dt = \frac{T}{3}$$

First basis function $\phi_1(t) = \dfrac{s_1(t)}{\sqrt{E_1}} = \sqrt{3/T} \;\; for\; 0 \le t \le T/3$

Step-2: Coefficient $s_{21}$

$$s_{21} = \int_0^T s_2(t)\phi_1(t)\, dt = \sqrt{T/3}$$

Energy of $s_2(t)$

$$E_2 = \int_0^T s_2^2(t)\, dt = \frac{2T}{3}$$

Second Basis function $\phi_2(t) = \dfrac{s_2(t) - s_{21}\phi_1(t)}{\sqrt{E_2 - s_{21}^2}} = \sqrt{3/T} \;\; for \;\; T/3 \le t \le 2T/3$

Step-3: Coefficient $s_{31}$:

$$s_{31} = \int_0^T s_3(t)\,\phi_1(t)\,dt = 0$$

Coefficient $s_{32}$

$$s_{32} = \int_0^T s_3(t)\,\phi_2(t)\,dt = \sqrt{T/3}$$

Intermediate function

$$g_3(t) = s_3(t) - s_{31}\Phi_1(t) - s_{32}\Phi_2(t)$$

$$g_3(t) = 1 \quad \text{for } 2T/3 \leq t \leq T/3$$

Third Basis function

$$\phi_3(t) = \frac{g_3(t)}{\sqrt{\int_0^T g_3^2(t)\,dt}} = \sqrt{3/T} \quad for \ 2T/3 \leq t \leq T$$

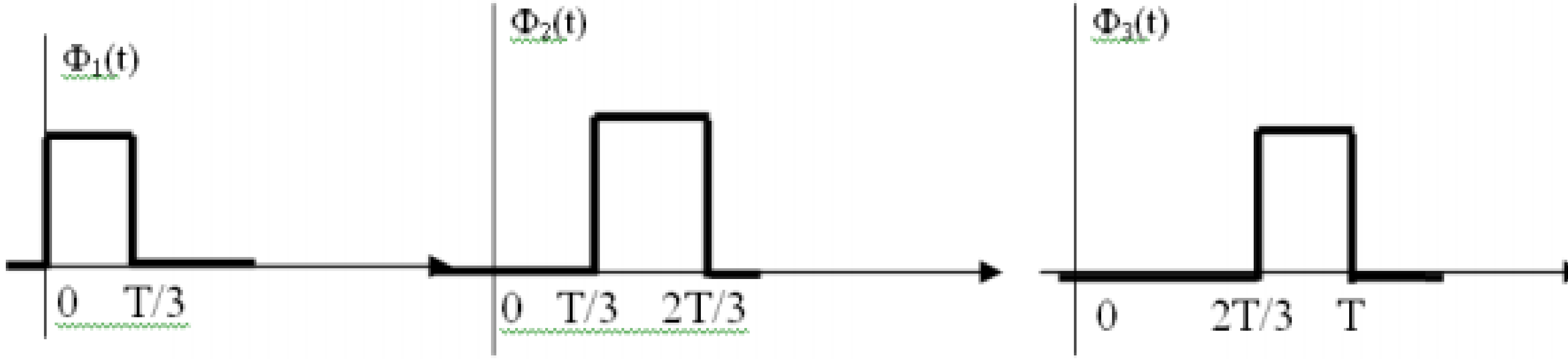



*Fig-P1.2: Orthonormal functions for the Problem-1*

Representation of the signals

$$S_1(t) = \sqrt{T/3}\ \phi_1(t)$$

$$\phi_3(t) = \frac{g_3(t)}{\sqrt{\int_0^T g_3^2(t)\,dt}} = \sqrt{3/T}\ \ for$$

$$\phi_1(t) = \frac{s_1(t)}{\sqrt{E_1}} = \sqrt{3/T}$$

$$\phi_2(t) = \frac{s_2(t) - s_{21}\phi_1(t)}{\sqrt{E_2 - s_{21}^2}} = \sqrt{3/T}$$

$$S_2(t) = \sqrt{T/3}\,\phi_1(t) + \sqrt{T/3}\,\phi_2(t)$$

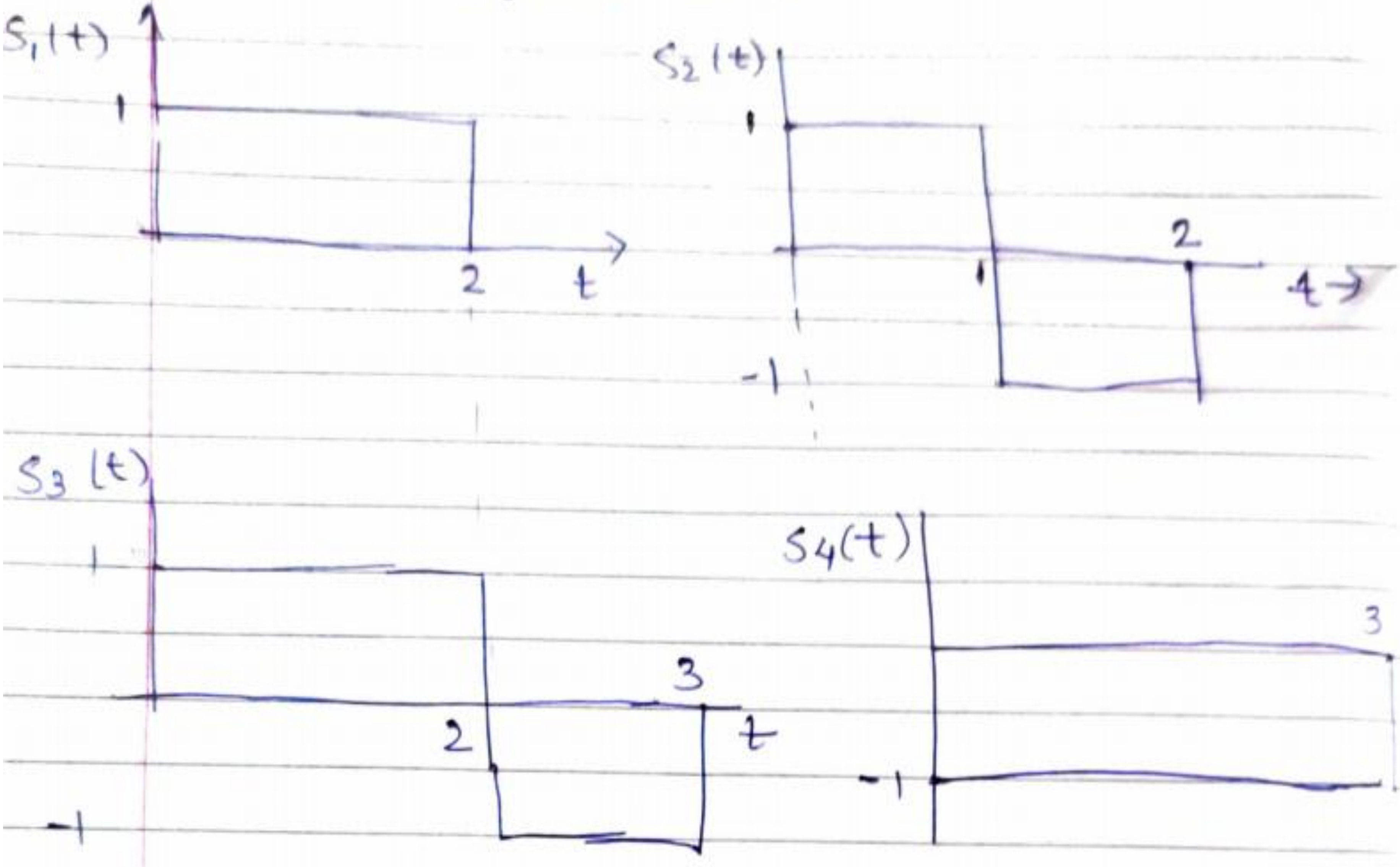
$$S_3(t) = \sqrt{T/3}\,\phi_2(t) + \sqrt{T/3}\,\phi_3(t)$$

$$S_4(t) = \sqrt{T/3}\,\phi_1(t) + \sqrt{T/3}\,\phi_2(t) + \sqrt{T/3}\,\phi_3(t)$$

- Each signal in the set $s_i(t)$ is completely determined by the vector of its coefficients

$$\mathrm{s_i} = \begin{bmatrix} s_{i1} \\ s_{i2} \\ . \\ . \\ . \\ s_{iN} \end{bmatrix}, \qquad i = 1,2,\ldots.,\mathrm{M} \qquad (5.8)$$

# Find Orthogonal Basis

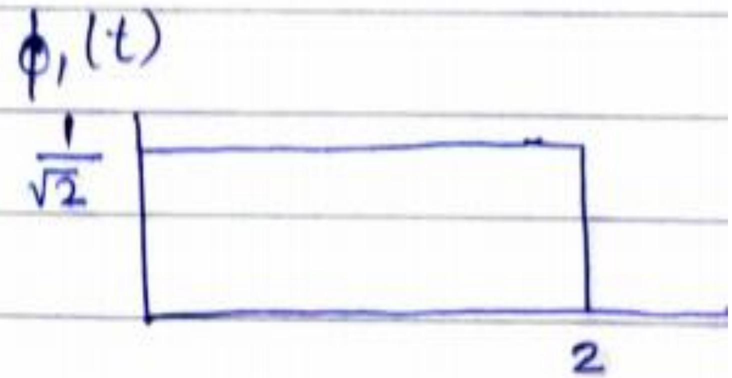


- Maximum orthogonal basis=?

$$\phi_1(t) = \frac{s_1(t)}{\sqrt{\mathcal{E}_1}} \qquad \mathcal{E}_1 = \int_{-\infty}^{\infty} |s_1(t)|^2 \, dt$$

$$\int_{-\infty}^{\infty} |1|^2 \, dt = \int_{0}^{2} dt = 2$$

$$\therefore \quad \phi_1(t) = \frac{s_1(t)}{\sqrt{2}}$$



Second basis function

$$\phi_2(t) = \frac{s_2(t) - s_{21}\,\phi_1(t)}{\sqrt{E_2 - s_{21}^2}}$$

$$E_2 = \text{Energy of } s_2(t) = \int_0^T s_2^2(t) =$$

$$= \int_0^1 1\,dt + \int_1^2 1\,dt = 1 + 1 = 2$$

Coefficient $s_{21}$

$$s_{21} = \int_0^T s_2(t)\,\phi_1(t)\,dt$$

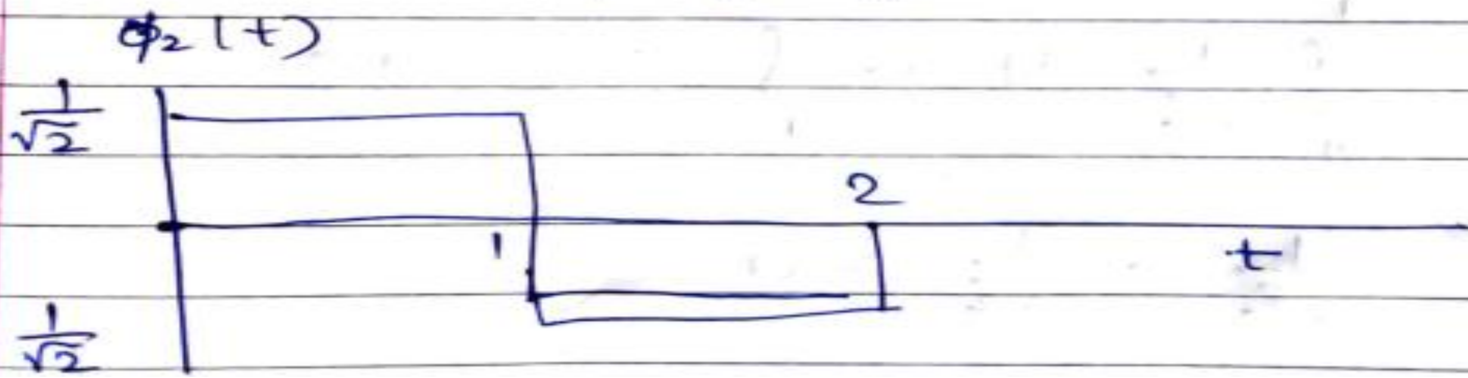
$$S_{21} = \int_0^T s_2(t)\,\phi_1(t)\,dt$$

$$= 0$$

$$\int_0^1 \frac{1}{\sqrt{2}}\,dt + \int_1^2 \left(-\frac{1}{\sqrt{2}}\right)dt = 0$$

$$\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}$$

$$\phi_2(t) = \frac{s_2(t)}{\sqrt{E_2 - S_{21}^2}} = \frac{s_2(t)}{\sqrt{2}}$$



$\phi_3(t)$

Coefficient $s_{31} = \int_0^T s_3(t)\, \phi_1(t)\, dt$

Coefficient $s_{32} = \int_0^T s_3(t)\, \phi_2(t)\, dt$ —

Intermediate function

$$g_3(t) = s_3(t) - s_{31}\phi_1(t) - s_{32}\phi_2(t)$$

$$\phi_3(t) = \frac{g_3(t)}{\sqrt{\int_0^T g_3^2(t)\, dt}}$$

$$S_{31} = \int_0^2 \left|\left(\frac{1}{\sqrt{2}}\right)\right| dt = \frac{1}{\sqrt{2}}\, 2 = \sqrt{2}$$

$$S_{32} = \int_0^1 \frac{1}{\sqrt{2}}\, dt + \int_1^2 \left(-\frac{1}{\sqrt{2}}\right) dt$$

$$= \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0$$

$$g_3(t) = s_3(t) - s_{31}\phi_1(t) - \underline{\underline{0}}$$

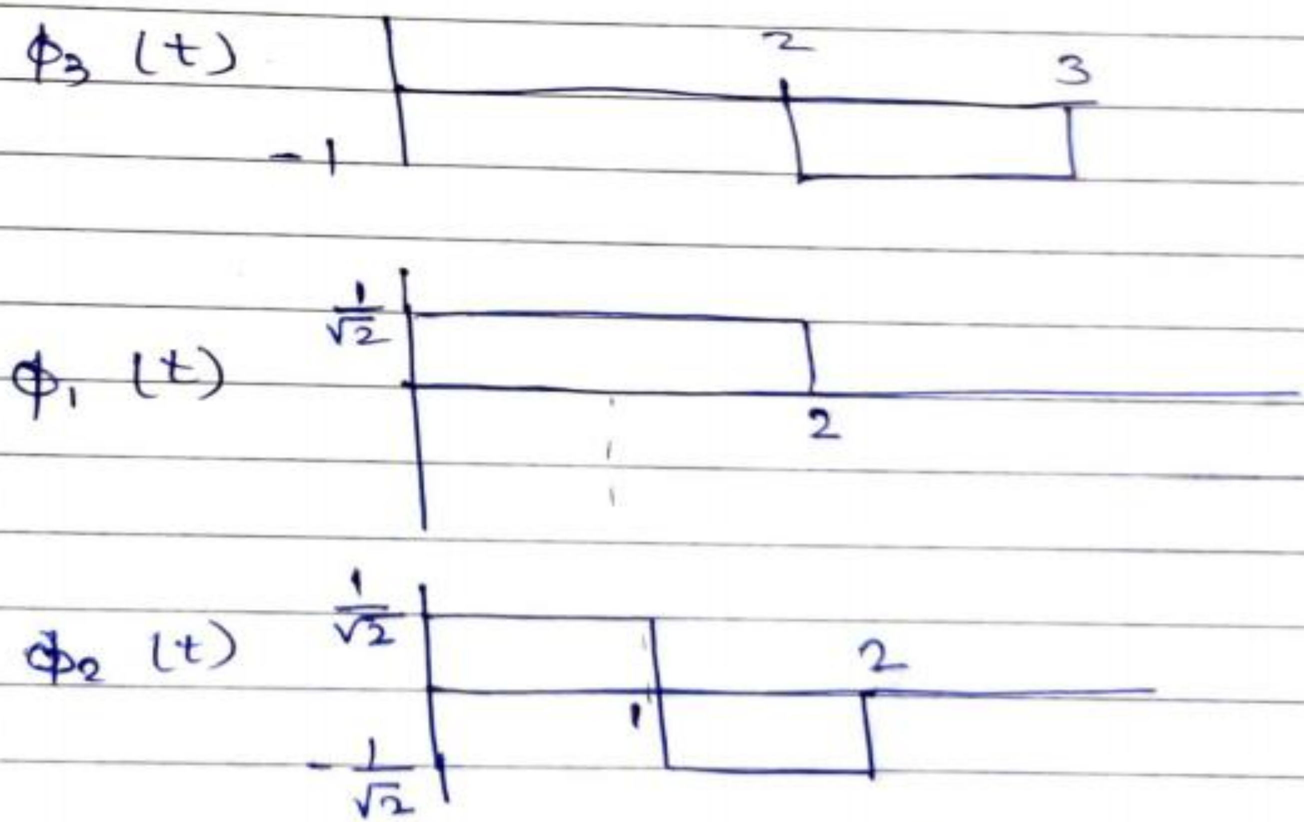
$$= s_3(t) - \sqrt{2}\,\phi_1(t)$$

$= 0$ otherwise

$= -1$ $2 \le t \le 3$

$$\phi_3(t) = \frac{g_3(t)}{\sqrt{\mathcal{E}_3}}$$

$$\mathcal{E}_3 = \int_{-\infty}^{\infty} |g_3(t)|^2\, dt = \int_2^3 (-1)^2\, dt = 1$$




$$g_4(t) = s_4(t) - s_{41}\phi_1(t) - s_{42}\phi_2(t) - s_{43}\phi_3($$

- The signal vector $\boldsymbol{s_i}$ concept can be extended to 2D, 3D etc. N-dimensional Euclidian space
- Provides mathematical basis for the geometric representation of energy signals that is used in noise analysis
- Allows definition of
  - Length of vectors (absolute value)
  - Angles between vectors
  - Squared value (inner product of $s_i$ with itself)

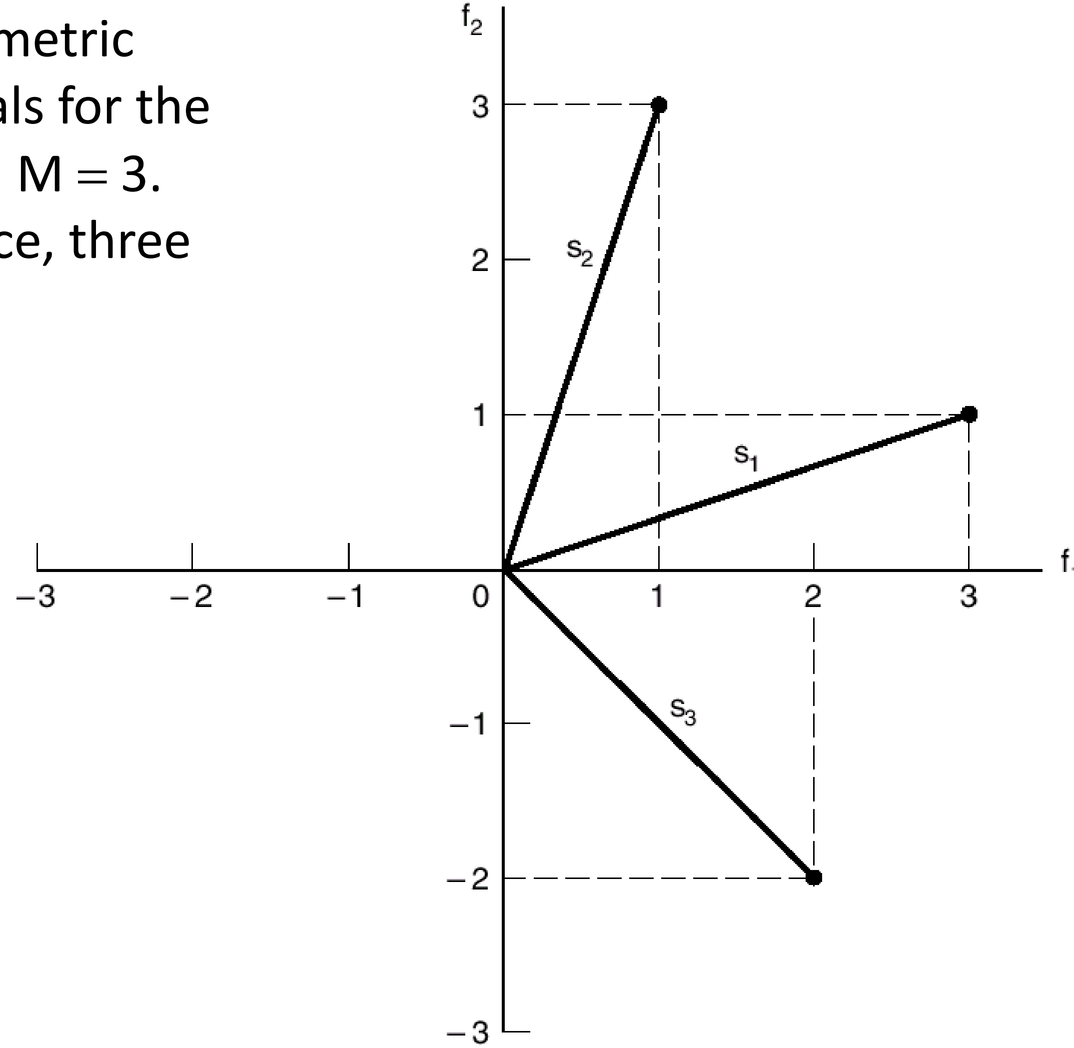
$$\|\mathbf{s}_{\mathrm{i}}\|^2 = s_i^T \mathbf{s}_{\mathrm{i}}$$

Matrix Transposition

$$= \sum_{j=1}^{N} s_{ij}^2, \qquad i = 1,2,\ldots,M$$

# Figure 5.4

Illustrating the geometric representation of signals for the case when N = 2 and M = 3. (two dimensional space, three signals)



# Also,

What is the relation between the *vector representation* of a signal and its *energy value?*

- …start with the definition of average energy in a signal…(5.10)

$$\mathrm{E_i} = \int_0^T s_i^2(t)dt$$

- Where $s_i(t)$ is as in (5.5):

$$s_i(t) = \sum_{j=1}^{N} s_{ij}\phi_j(t),$$

- After substitution: $$E_i = \int_0^T \left[ \sum_{j=1}^{N} s_{ij}\phi_j(t) \right] \left[ \sum_{k=1}^{N} s_{ik}\phi_k(t) \right] dt$$

- After regrouping: $$E_i = \sum_{j=1}^{N} \sum_{k=1}^{N} s_{ij} s_{ik} \int_0^T \phi_j(t)\phi_k(t)\,dt$$

- $\Phi_j(t)$ is orthogonal, so finally we have: $$E_i = \sum_{j=1}^{N} s_{ij}^2 = \|s_i\|^2$$

The *energy of a signal* is equal to the *squared length of its vector*

# Formulas for two signals

- Assume we have a pair of signals: $s_i(t)$ and $s_j(t)$, each represented by its vector,
- Then:

$$s_{ij} = \int_0^T s_i(t)s_k(t)dt = s_i^T \mathrm{s}_\mathrm{k}$$

Inner product of the signals is equal to the inner product of their vector representations [0,T]

Inner product is invariant to the selection of basis functions

# Euclidian Distance

- The Euclidean distance between two points represented by vectors (signal vectors) is equal to $||s_i - s_k||$ and the squared value is given by:

$$\|\mathrm{s}_\mathrm{i} - \mathrm{s}_k\|^2 = \sum_{j=1}^{N} (s_{ij} - s_{kj})^2$$

$$= \int_0^T (s_i(t) - s_k(t))^2 \, dt$$

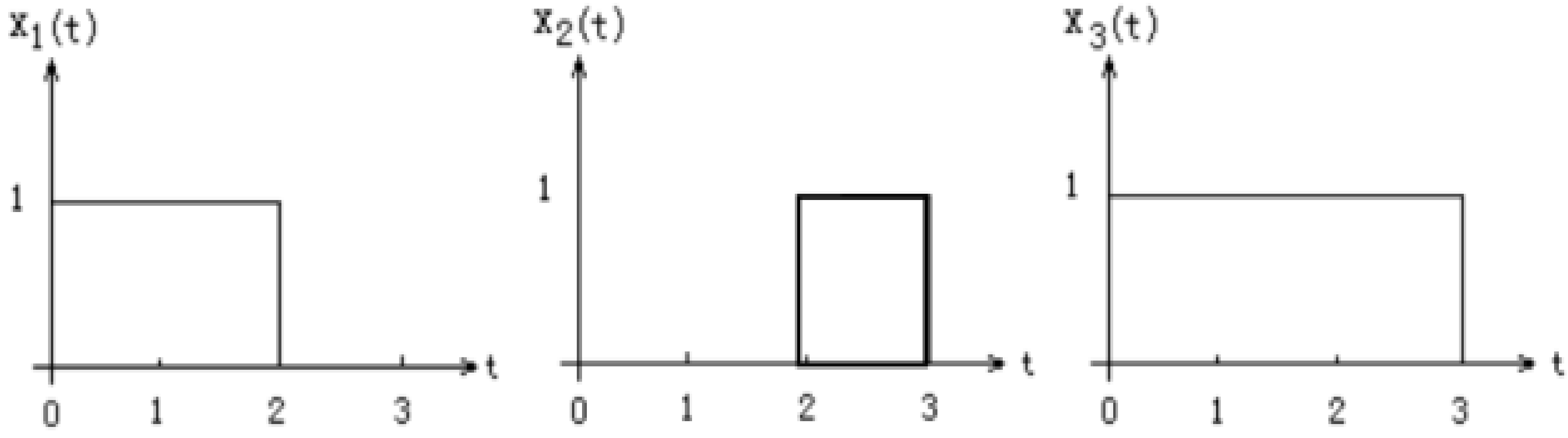
# Angle between two signals

- The *cosine of the angle $\Theta_{ik}$* between two signal vectors $s_i$ and $s_k$ is equal to the inner product of these two vectors, divided by the product of their norms:

$$\cos\theta_{\mathrm{ik}} = \frac{s_i^T \mathrm{s}_{\mathrm{k}}}{\|s_i\| \|s_k\|}$$

- So the two signal vectors are *orthogonal* if their inner product $s_i^T s_k$ is zero ($\cos \Theta_{ik} = 0$)

Use the Gram-Schmidt procedure to find a set of orthonormal basis functions corresponding to the signals shown below




Express $x_1$, $x_2$, $x_3$, in terms of the orthonormal basis functions found previously

Draw the constellation diagram for this signal set

# Question

- How can you determine the dimension of a signal space? via Gram-Schmidt
- What does the projection theorem state? projection error is orthogonal to subspace
- Explain the Gram-Schmidt procedure?
  1. Choose one vector, normalize it $\rightarrow$ first basis.
  2. Project next vector onto current basis. If projection error is zero, no new basis. Otherwise normalize projection error (orthogonal to current basis) $\rightarrow$ next basis.
  3. Repeat step 2 for all remaining vectors.